

TUSHAR KHITOLIYA

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EDUCATION

B.Tech in Electrical and Electronics Engineering

NIT- Delhi, 2021 – 2025

Intermediate (non-medical)

Sarvodaya Vidyalaya, Delhi

Matriculation

DAV Public School, Delhi

TECHNICAL SKILLS

- **AI/ ML** : NumPy, Pandas, Scikit learn, TensorFlow, Pytorch, LLMs, RAG, Deep Learning, NLP, Computer Vision
- **Machine Learning**: Supervised Learning, Unsupervised Learning, Regression, Classification, Clustering, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Scikit-learn
- **Tools**: GitHub, VS Code, Jupyter Notebook, Streamlit, Tableau, Plotly, Matplotlib, Seaborn, Hugging face, Lang chain
- **Languages**: Python, SQL, HTML, CSS

PROJECTS

1. YouTube Video RAG Chatbot

- Built a Retrieval-Augmented Generation chatbot in Flask to answer natural-language questions grounded in indexed YouTube video transcripts.
- Implemented semantic search using LangChain and Pinecone with multilingual-e5-large embeddings and MMR retrieval, tuning top-k and diversity parameters ($k=10$, $\lambda=0.2$) for relevant, non-redundant context.
- Designed a context-injection prompt pipeline to constrain LLM responses strictly to retrieved transcript content, minimizing hallucination.

2. Plant Disease Diagnosis

- Implemented an AI-driven image classification model using CNNs to accurately detect and diagnose plant diseases from leaf images with high precision.
- Integrated a chat-based interface for instant disease insights and prevention tips.

3. AI-Powered Multi-Document Conversational RAG System

- Built a scalable RAG pipeline enabling chat over multiple documents (100+ pages) across PDFs, images, text.
- Improved document retrieval speed by 70–80% using FAISS-based semantic search
- Increased response relevance by 30% with conversational memory for context-aware answers Intelligent Invoice Extraction & QA System.

4. Learning Management System (LMS) – Full Stack Web Application

- Built a full-stack LMS using Flask, MySQL, and SQLAlchemy with features like student enrollment, batch management, and admin controls.
- Implemented dynamic search and filtering, improving student data retrieval efficiency by 60%
- Managed 500+ student records with optimized database queries and scalable backend APIs.

PROFESSIONAL EXPERIENCE

2. Product Engineer and Instructor, Coding Blocks Pvt. Ltd.

May 2024- Present

- Delivered AI and ML training to undergraduate and graduate students, emphasizing practical learning and innovation. Mentored real-world projects and received excellent feedback for engaging outcome-driven teaching.
- Instructed students on real-world projects, fostering innovation and problem-solving skills.

3. Data Science Intern, Mahen Technology Pvt. Ltd.

Mar 2026- Apr 2023

- Built and deployed ML models for text analysis, optimizing accuracy through effective data preprocessing and feature engineering.
- Extracted and analyzed datasets using Python to perform advanced text analysis.
- Optimized accuracy through data preprocessing and feature engineering.

4. Data Science Intern, ExposysData Labs

Jan 2026- Feb 2026

- Delivered actionable insights by visualizing data trends, enhancing business understanding of customer demographics.
- Implemented customer segmentation using K- Means Clustering, analyzing customer data to identify age and salary patterns for targeted insights.

CERTIFICATIONS

- **Microsoft Certified : Azure AI Fundamentals (AI-900) – Microsoft**
- Certificate in **Master Data Science and Machine Learning – CodingBlocks**
- **Generative AI Applications with RAG and Lang chain-IBM**
- **Professional Certificate in SQL and SQL for Data Analysis – UDEMY**
- Introduction to **RAG – Duke University**
- Fundamentals of **AI Agents using RAG and Lang chain- IBM**